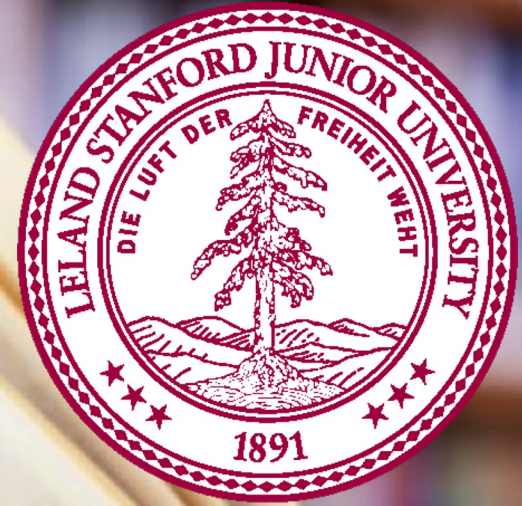




Lesson 1

Welcome to Code in Place

In this first video

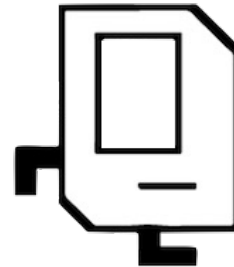


Meet the Team



What is Code in Place?

After: Learn to Code!



Meet the team!

Mehran Sahami



- Childhood: Iran
- High School: San Diego
- Stanford University Ph.D. in Machine Learning
(Before Machine Learning was cool)
- Spent a decade in tech industry before coming back as professor
 - Love of teaching is why I came back



Mehran Sahami



Piech and Sahami, Code in Place



Chris Piech

Teaching at Stanford

8,000+ students over 10 years

CS106A

Programming
Methodologies

CURRENT

CS106B

Programming
Abstractions

LAST: FALL 2016

CS109

Probability for Computer
Scientists

LAST: FALL 2018

CS221

Intro to Artificial
Intelligence

LAST: SUM 2013

Invented the
algorithm
behind bird
brain



Lead a research lab in:
Future of Education



Grew up in Nairobi, Kuala Lumpur before Stanford!

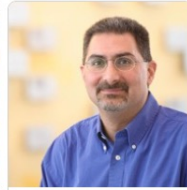
Piech and Sahami, Code in Place



What An Awesome Team



Chris Piech



Mehran
Sahami



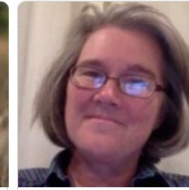
Ali Malik



Brahm Capoor



Juliette
Woodrow



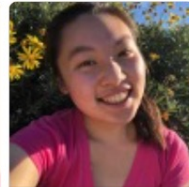
Julie Zelenski



Thomas
Jefferson



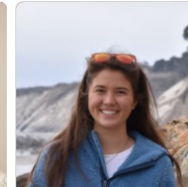
Miranda Li



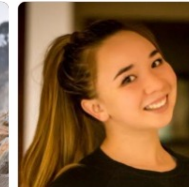
Patricia Wei



Cameron
Mohne



Sierra Wang



Hannah
Cussen



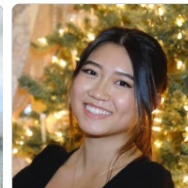
Jennifer Xu



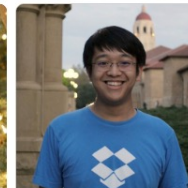
Joseph Tey



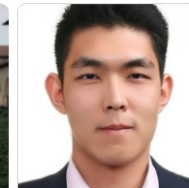
Julia Markel



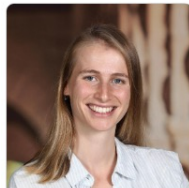
Sarah Jade Yao



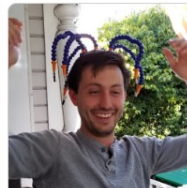
Allen Nie



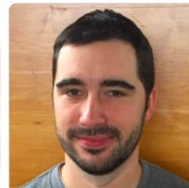
Yunsung Kim



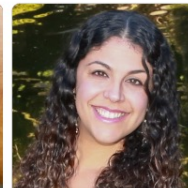
Dora Demszyk



Jason Ford



Jim Malamut



Jennifer
Langer-Osuna



Hannah Clay



Justin
Blumencranz

Piech and Sahami, Code in Place



It Takes a Village!

Aaron Broder, Abdu Mohamdy, Abdulwahab Omira, Adam Mosharrafa, Ahmet Uysal, Akinbowale Akin-Taylor, Akshaya Dinesh, Alan Cheng, Alejo, Alexandre Bucquet, Ali Malik, Amelie Byun, Ana Saavedra, Anand Shankar, Andrea Collins, Andrew Tierno, Areeba Khan, Arun Kulshreshtha, Ashley Taylor, Baker Sharp, Baris Bozkurt, Ben Barnett, Ben Holtz, Ben Ulmer, Cameron Mohne, Cameron Schaeffer, Caroline Tran, Cathy Zhang, Daniel Chao, Daniela Gonzalez, Dora Demszky, Elena Mosse, Eric Chew, Erin McCoy, Ethan Chi, Ethan Roy, Francisco Pernice, Frank Chen, Gonzalo Revuelta, Grace Hu, Griffin Dietz, Griffin Koontz, Guy Blanc, Ha Tran, Hannah Zhang, Haven Whitney, Heidi Chen, Inessa Roy, Irene Han, Isaac Pohl-Zaretsky, Jackson Eilers, Jacob Sharf, Jason Chen, Jason Ford, Jason Ma, Je-Mé Kruger-Baartjes, Jennie Yang, Jerry Chen, Jessie Duan, Jillian Tang, John Cho, Joseph Lee, Julia Chin, Julia Daniel, Julia Lee, Julia Truitt, Juliette Woodrow, Kaili Wang, Kartik Chandra, Kat Gregory, Kate Rydberg, Katherine Erdman, Kayla Patterson, Kevin Shin, Kurt Berglund, Kylie Jue, Lee Alpert, Lekan Wang, Lena Blackmon, Linda Tong, Lisa Einstein, Lisa Wang, Lisa Yan, Logan Marquis, Lucía M, Maneesh Apte, Matt King, Matt Peng, Maxwell Bigman, Maya Ziv, Michael Chang, Mike Wu, Miranda Li, Miroslav Suzara, Moussa Doumbouya, Nadin Tamer, Nancy Hoang, Natalie Cygan, Nathan Dalal, Neel Kishnani, Nick Bowman, Nick McKeown, Nikhil Raghuraman, Nishant, Parth Sarin, Patricia Wei, Rich Davis, Ryan Eberhardt, Ryan Lian, Sabri Eyuboglu, Sam King, Samantha Kim, Sasha Ronaghi, Sathya Edamadaka, Sauren Khosla, Scott Maxwell, Segev Tsfati, Sejal Jain, Shreya Shankar, Sierra Kaplan-Nelson, Sonja Johnson-Yu, Sophie Andrews, Stephanie Palocz, Swati Daub, Tague Griffith, Tara Balakrishnan, Tara Iyer, Toby Bell, Tori Qiu, Troy Shen, Tyler Yep, Tyrell Baker, Varun Tandon, Will Crichton, William Ellsworth, Yifei Men, Yosefa Gilon, Zachary Birnholz, Zack Cinquini, Zheng Lian



What is Code in Place?



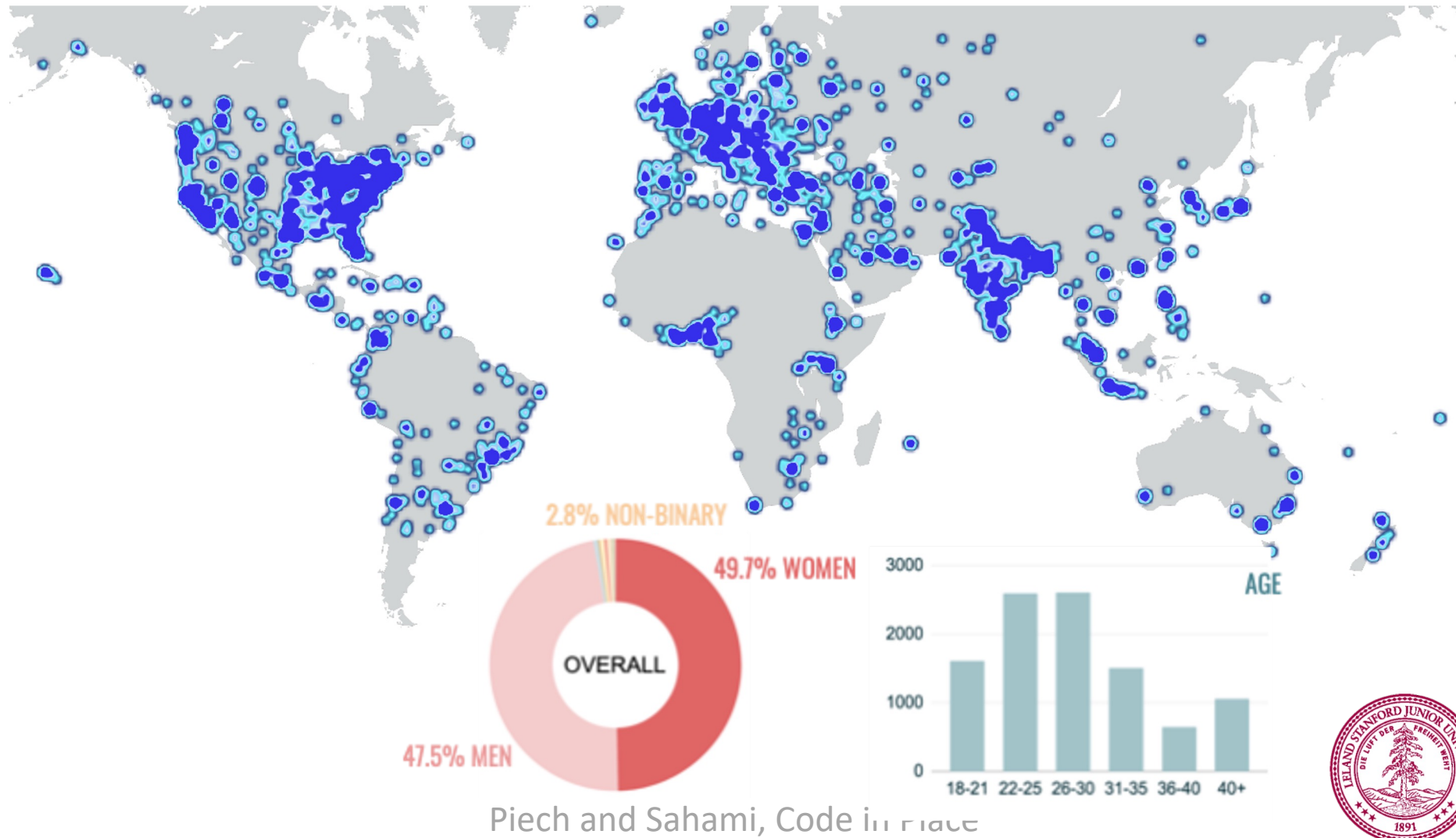
Code in Place:



1/2 of CS106A
As Community Service



Learners from over 150 countries
and all walks of life have completed this course



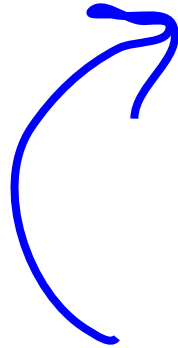
This class is just for **fun**

(what could be more fun than learning)

We are doing this as **volunteer**
work. You get a class for free!

Learning Goal

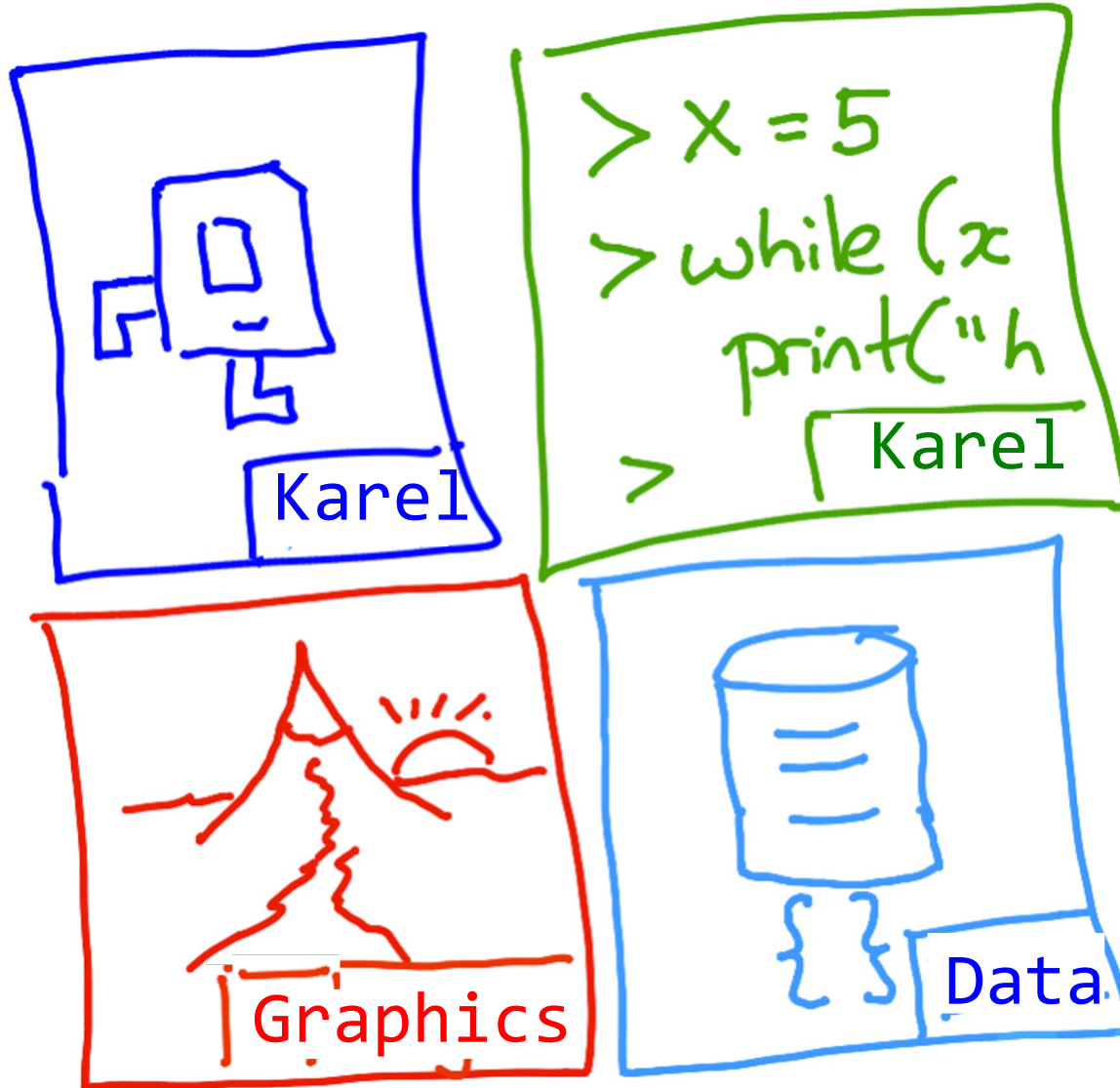
**Learn how to solve
problems with
python**



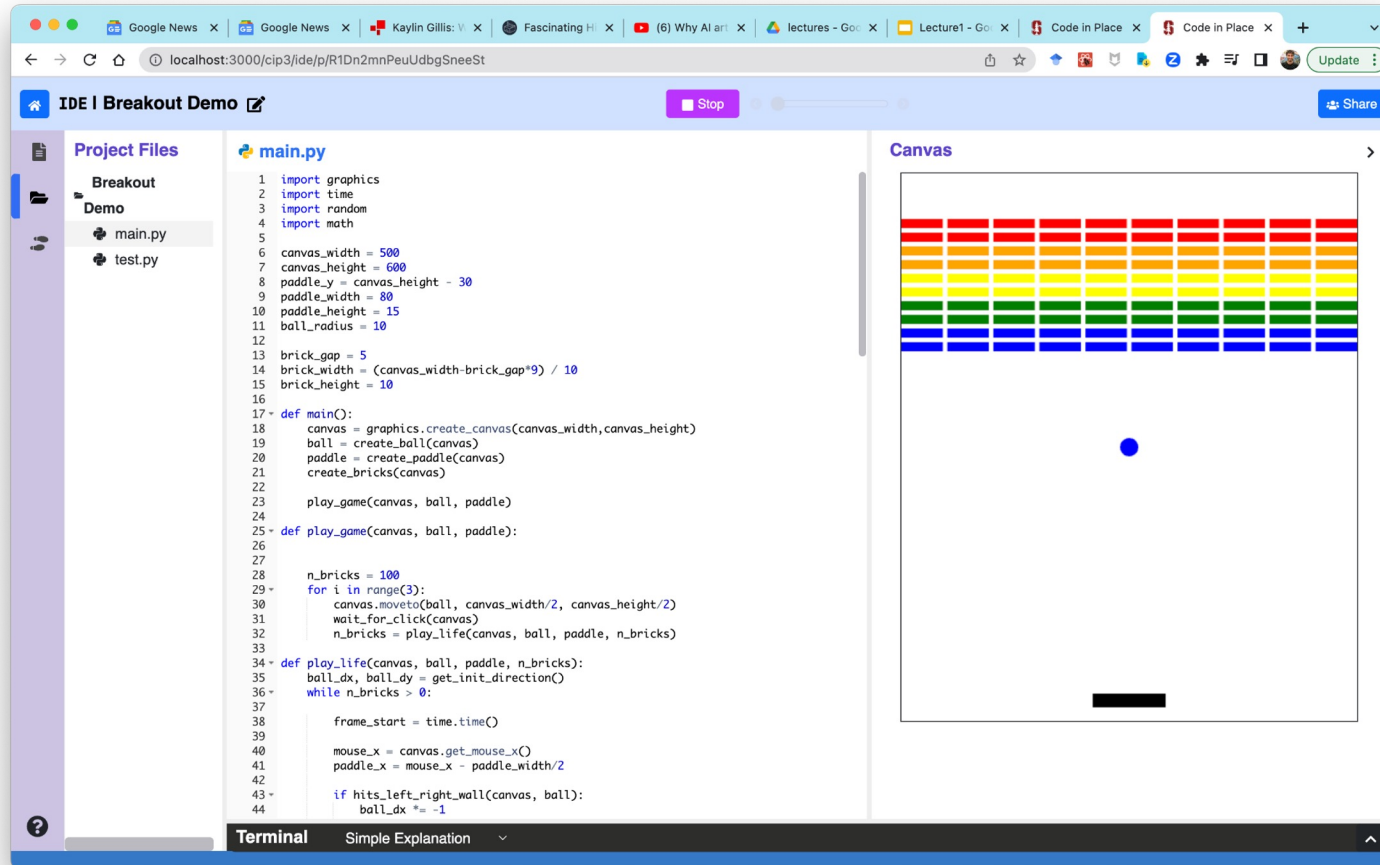
Find the joy
of a new skill



Course Plan



Programming Along the Way



Course Values

Art of Computer *Science*



Learn by Doing



Gratitude



Everyone is Welcome



We are all learners, teachers

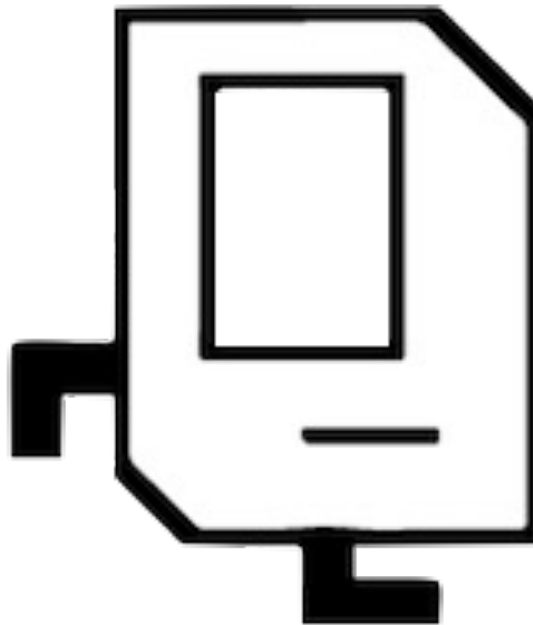


This is a collective
community service
project. You are part of
our community.

Help us make this work.



Let's learn!



Meet Karel the Robot

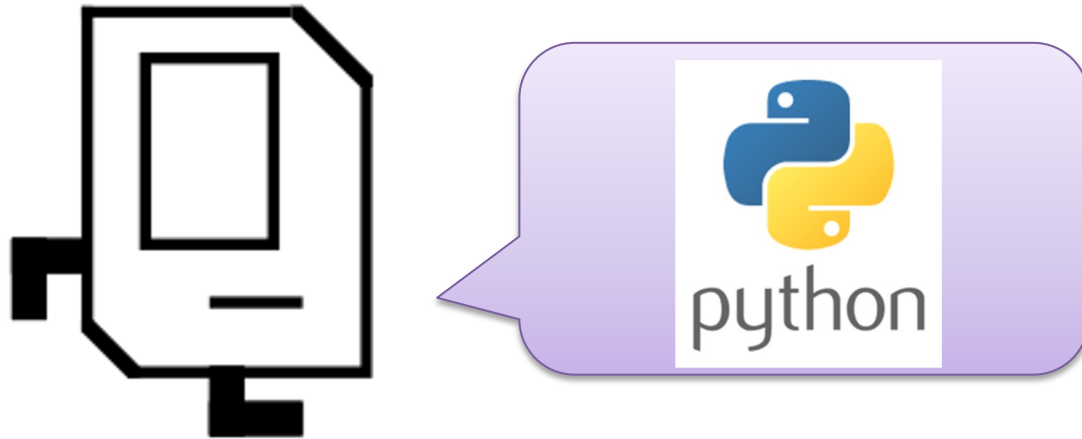
Rich Pattis



Good morning

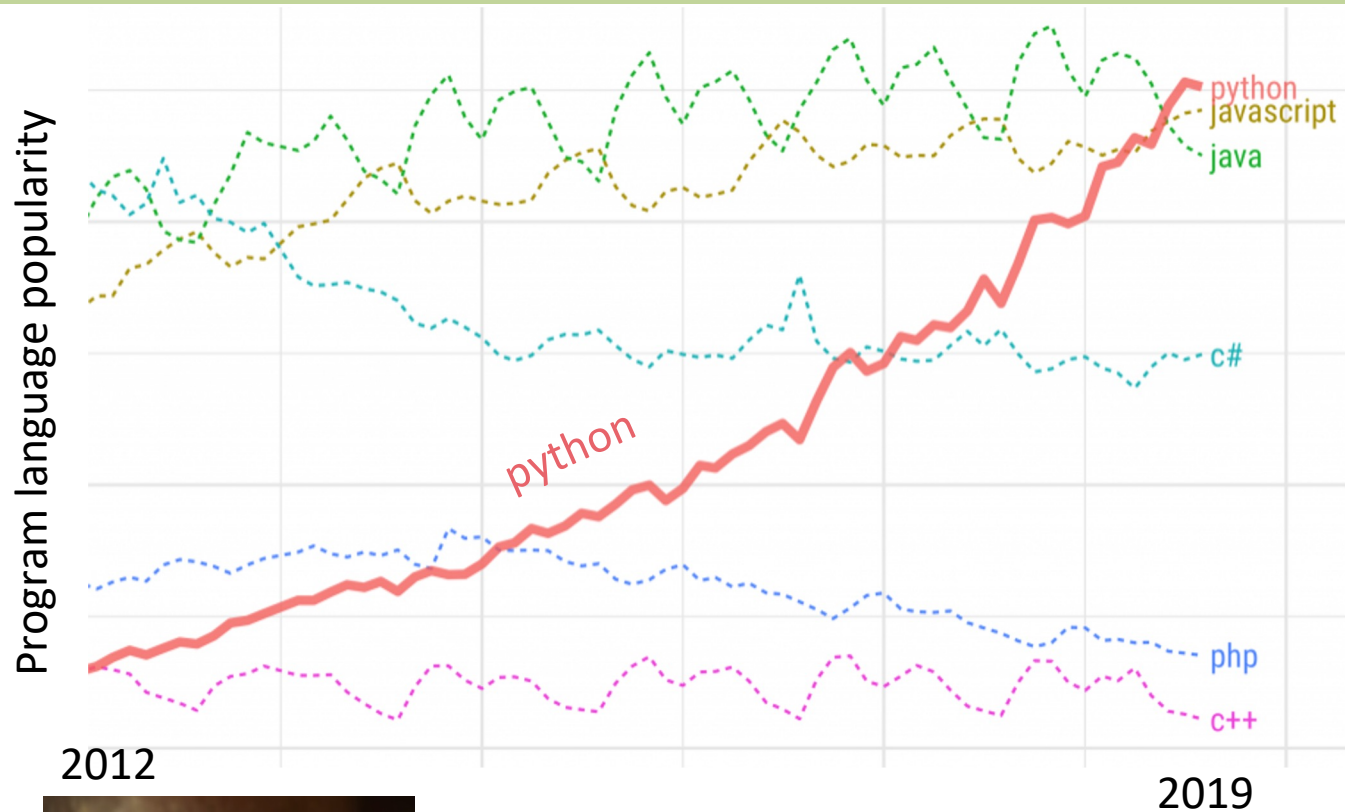


Karel Speaks Python



Why Python?

1



2

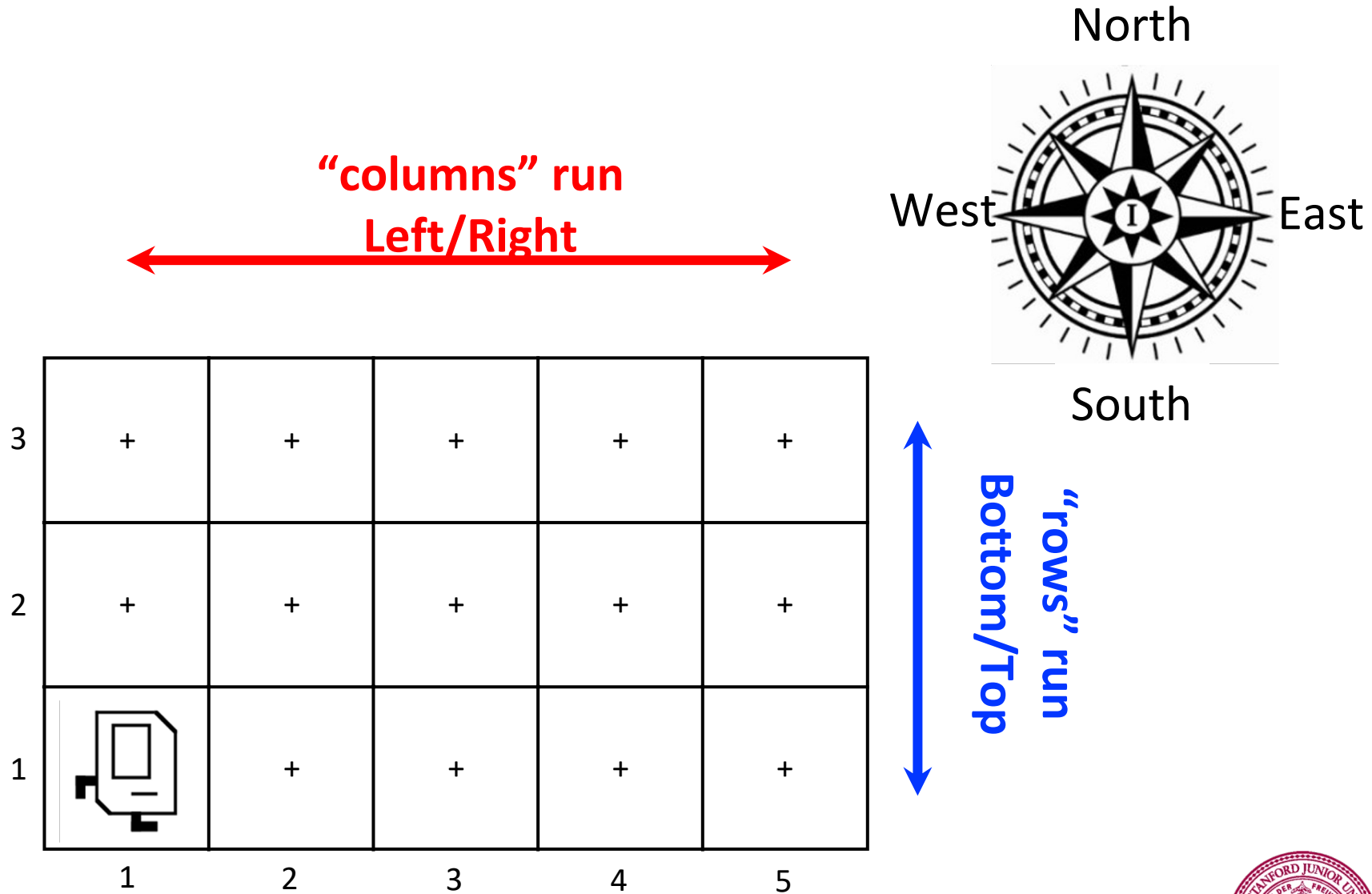


<https://stackoverflow.blog/2017/09/06/incredible-growth-python/>

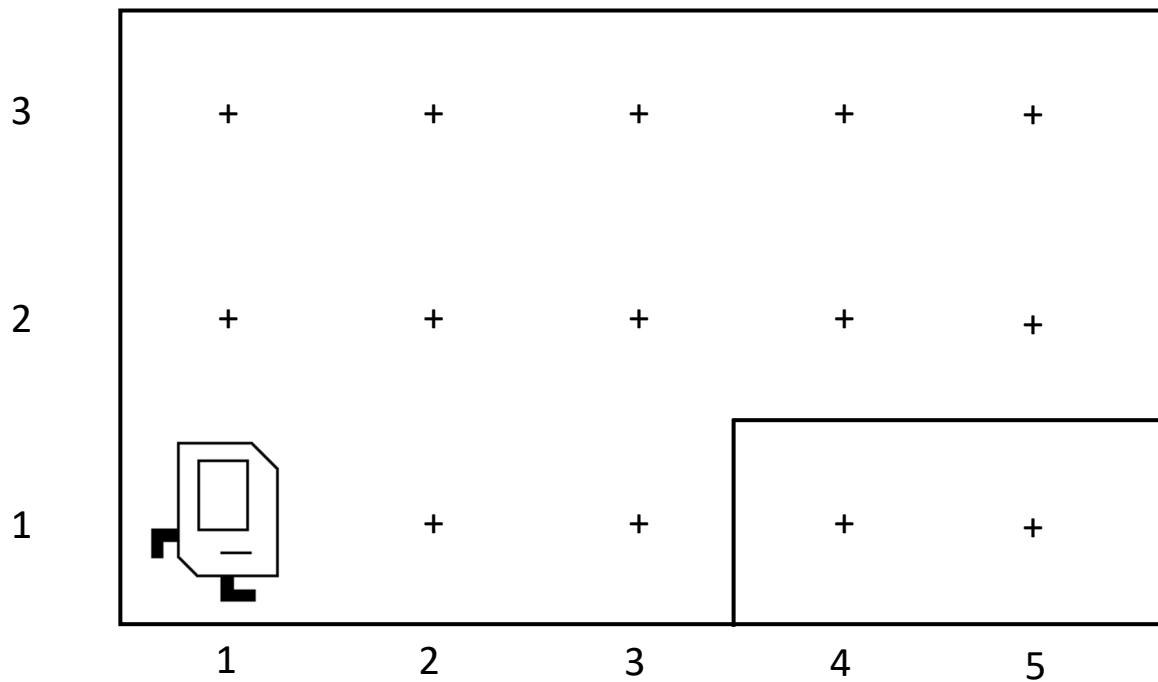
Guido van Rossum



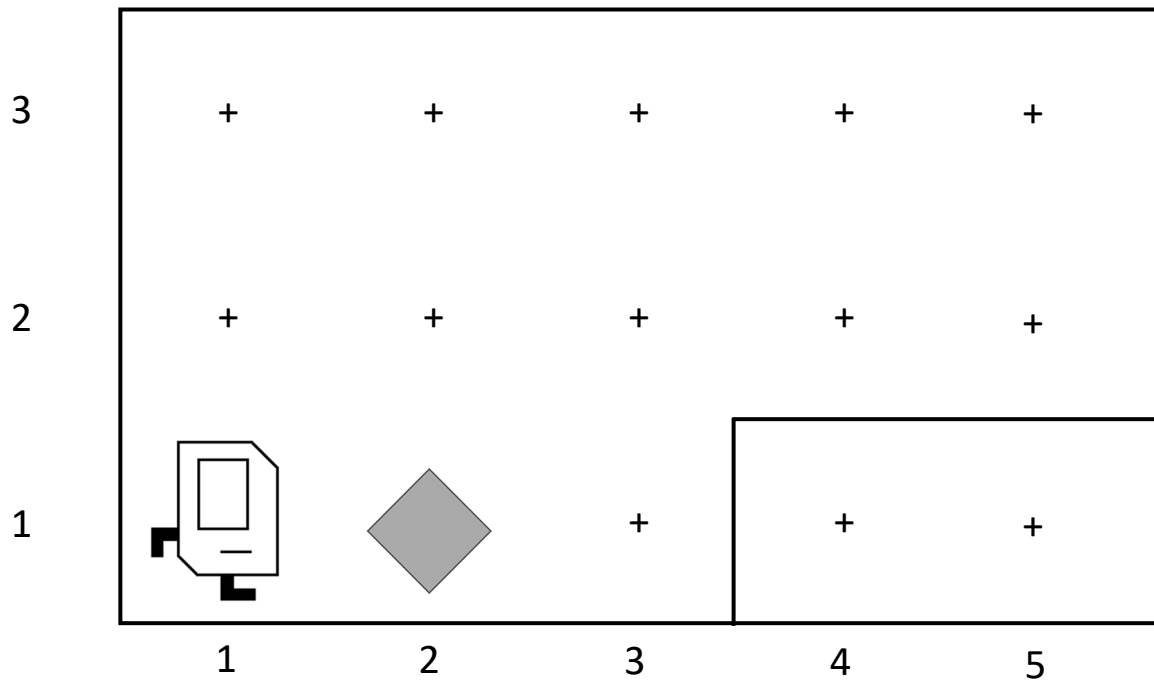
Karel's World



Walls



Beeper's



Knows Four Commands



`move()`

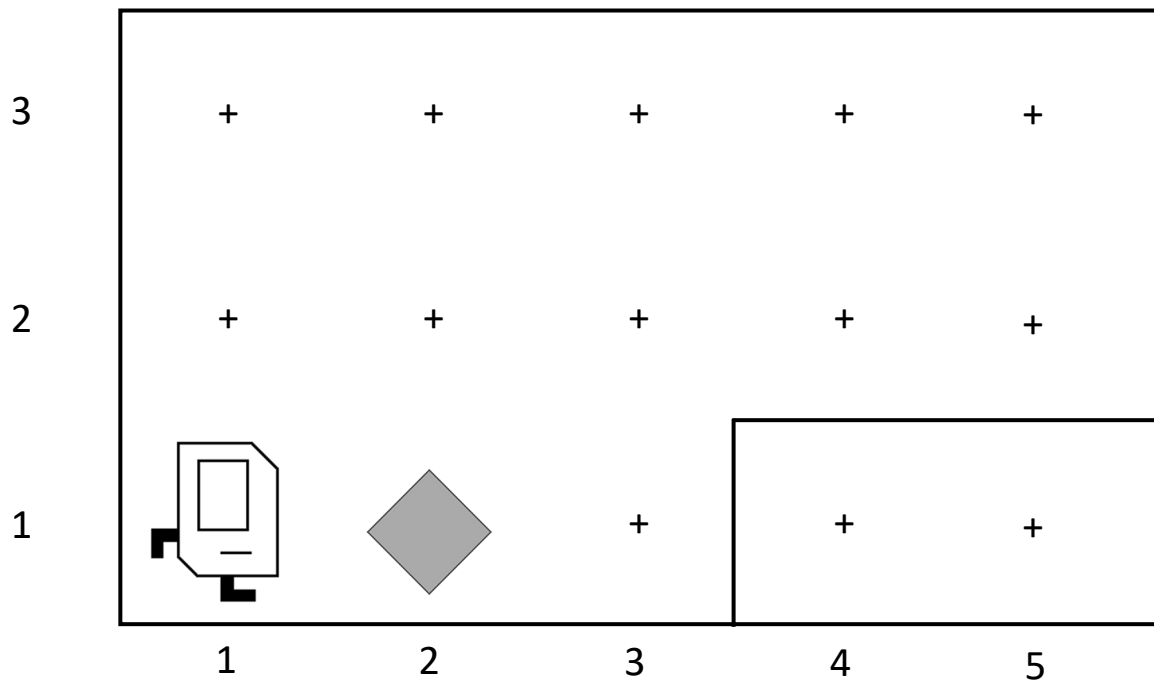
`turn_left()`

`put_beeper()`

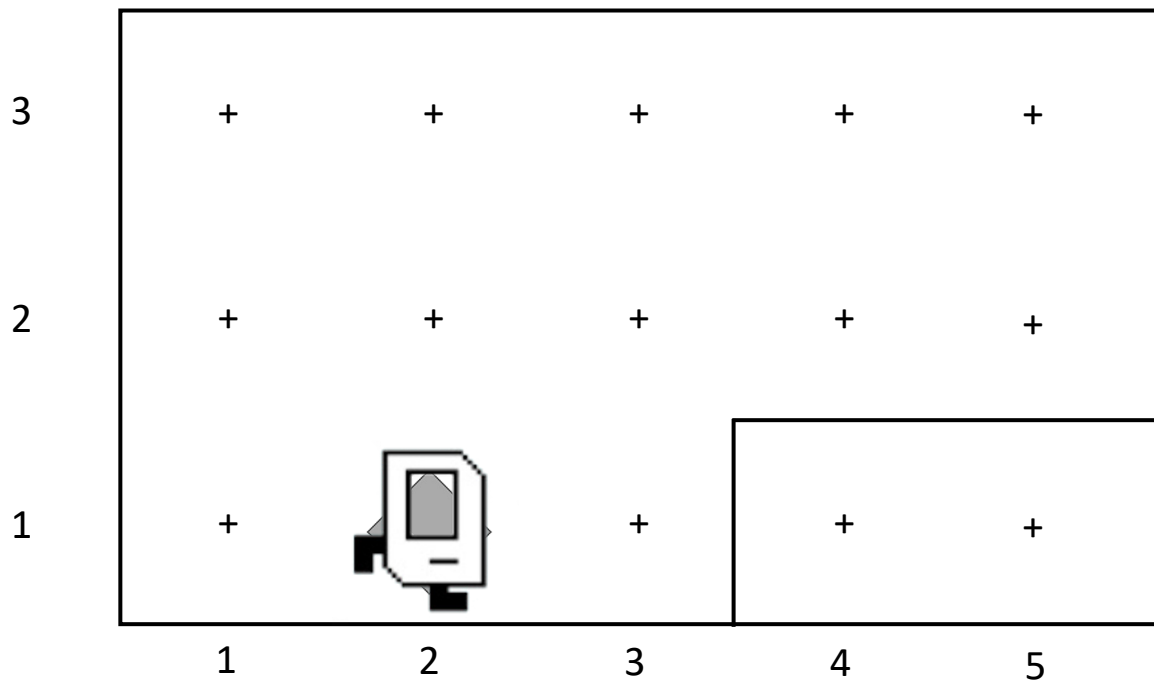
`pick_beeper()`

move ()

move ()

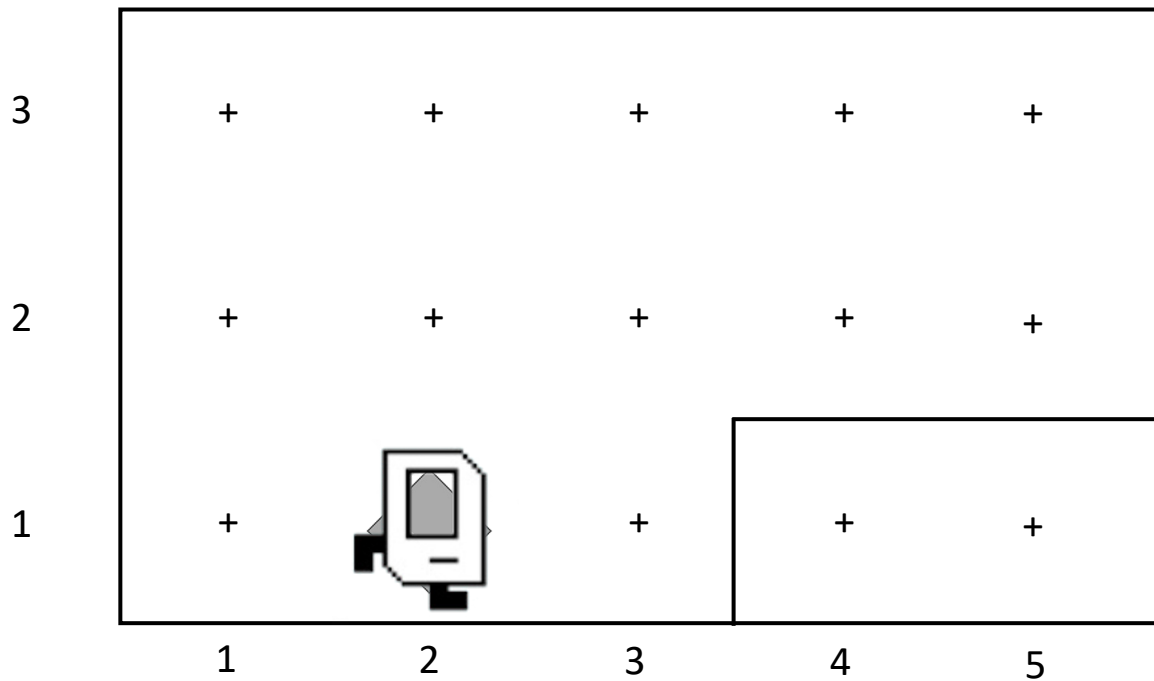


move ()

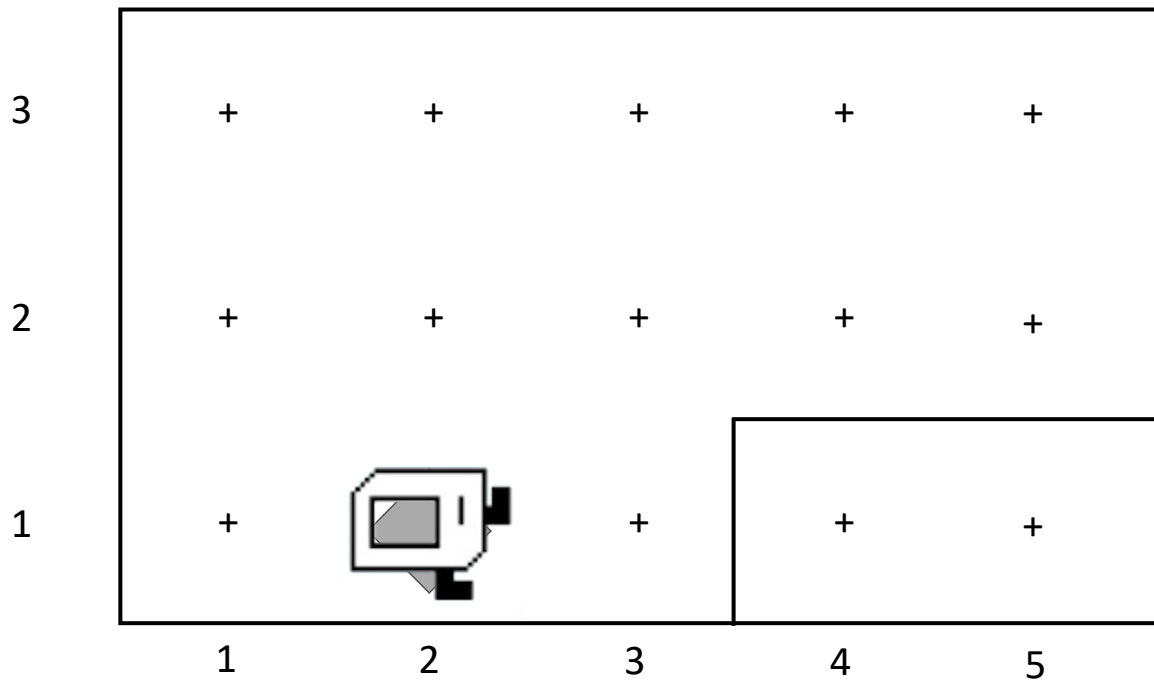


```
turn_left()
```

turn_left()

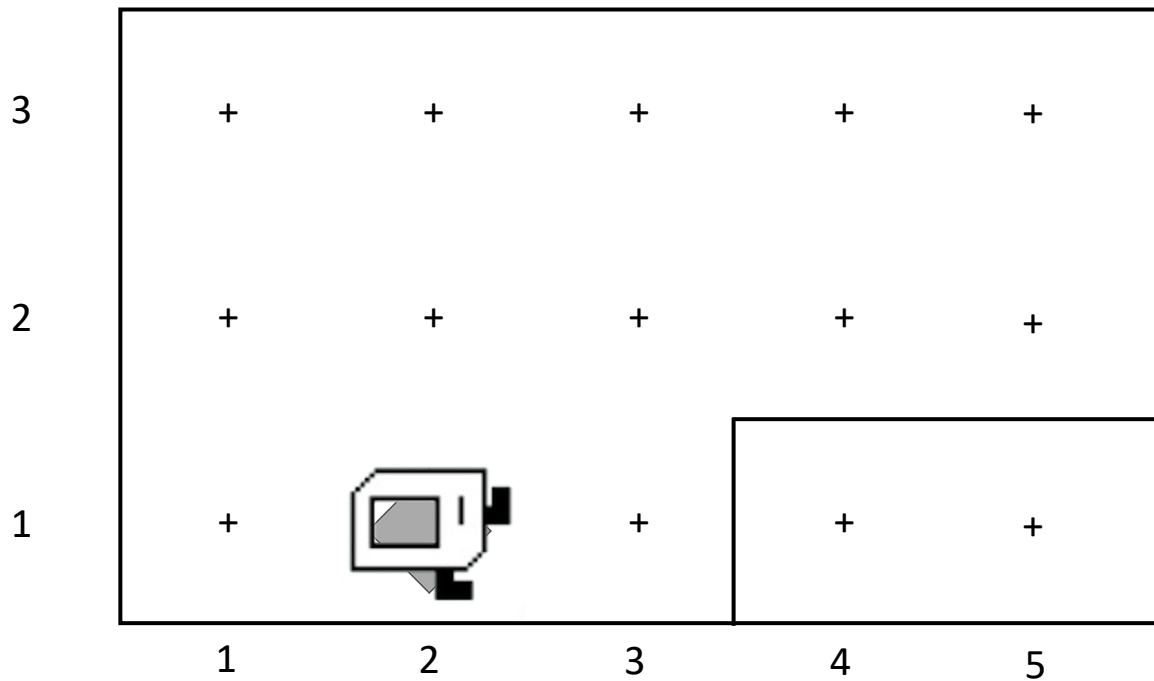


turn_left()

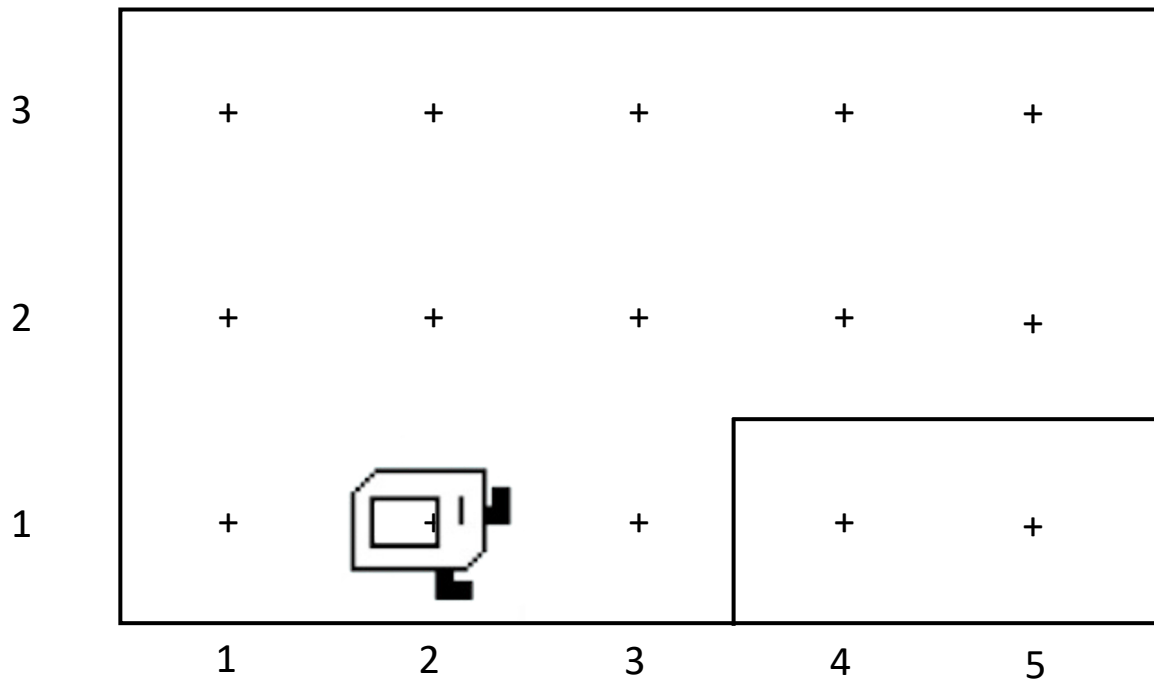


```
pick_beeper()
```

turn_left()

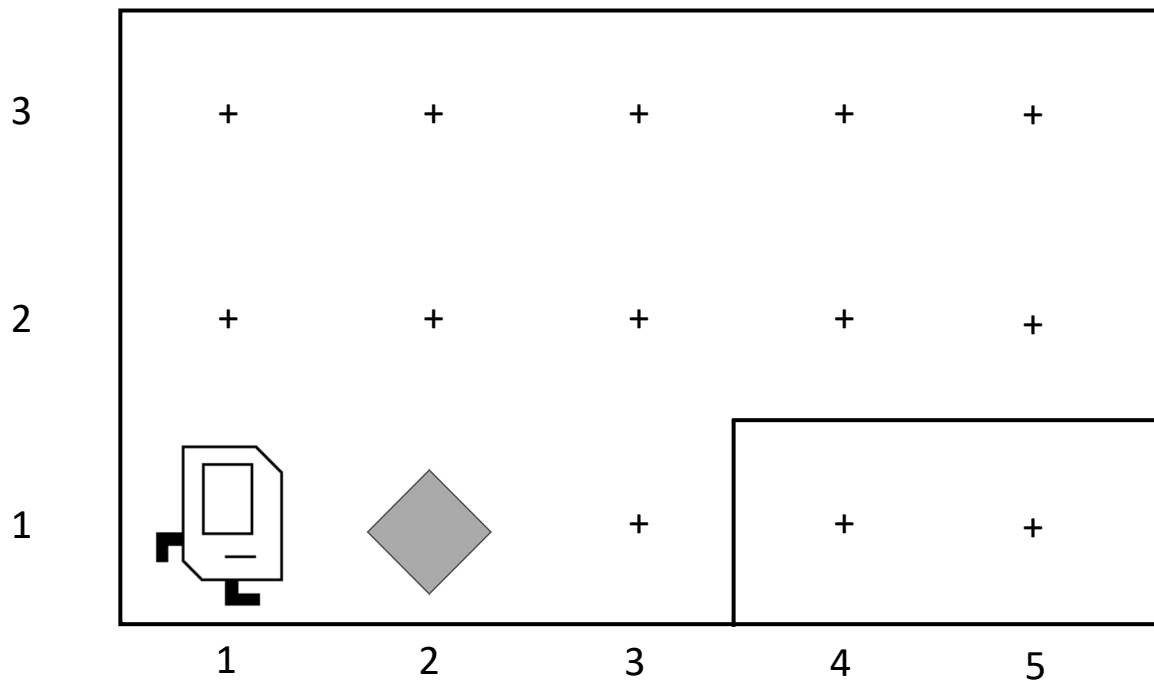


turn_left()

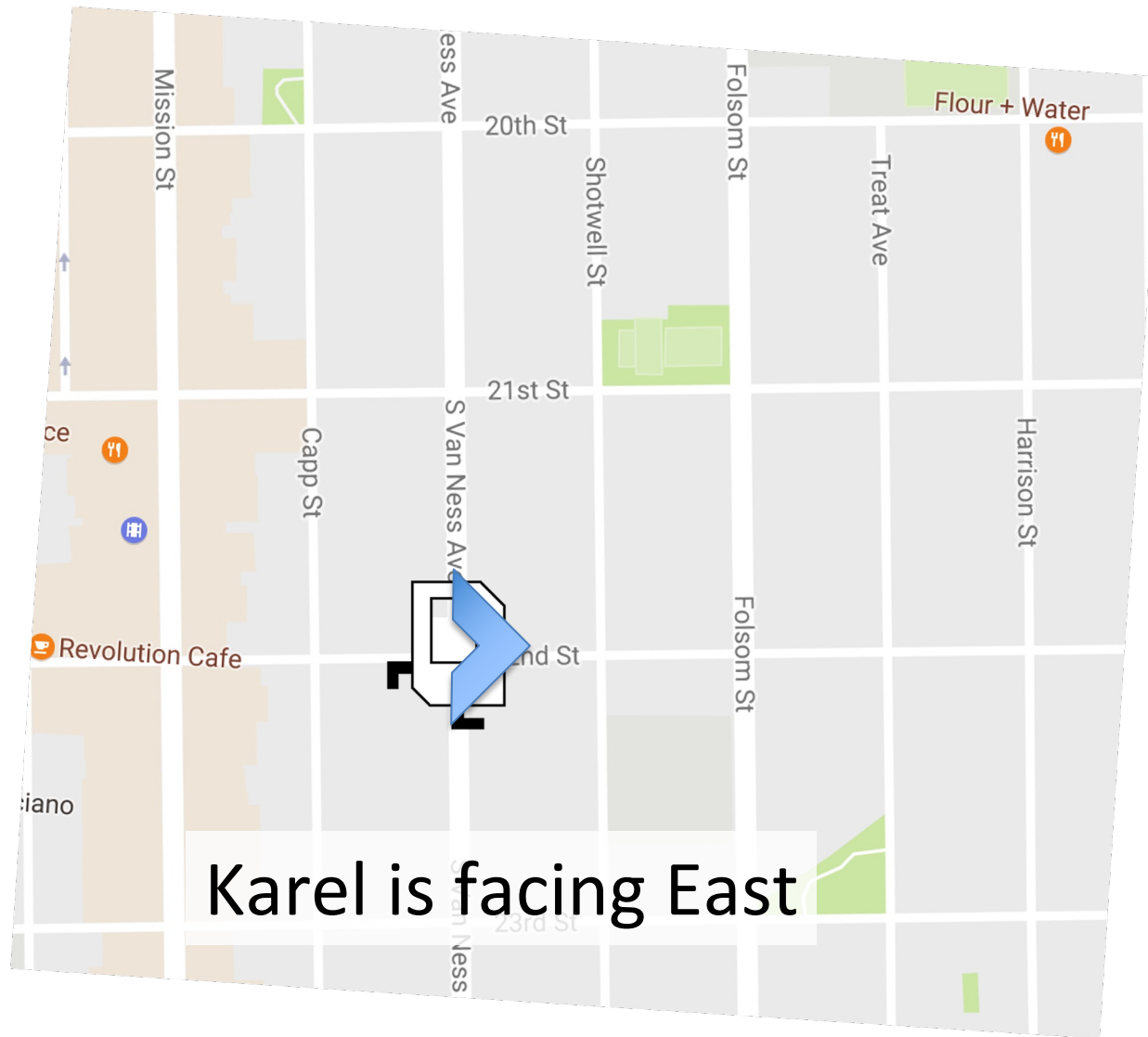


Make Sense?

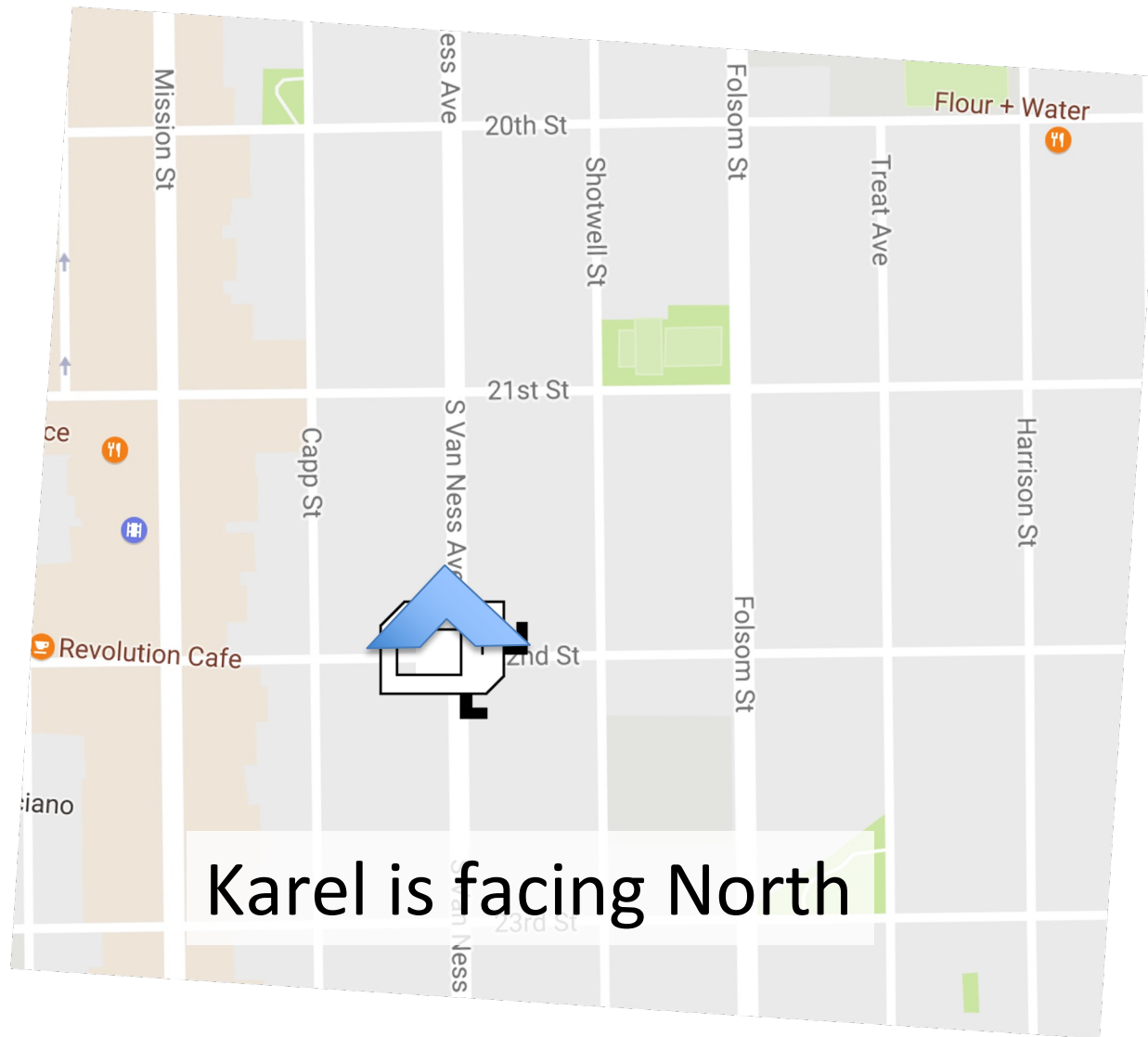
Bird's Eye View



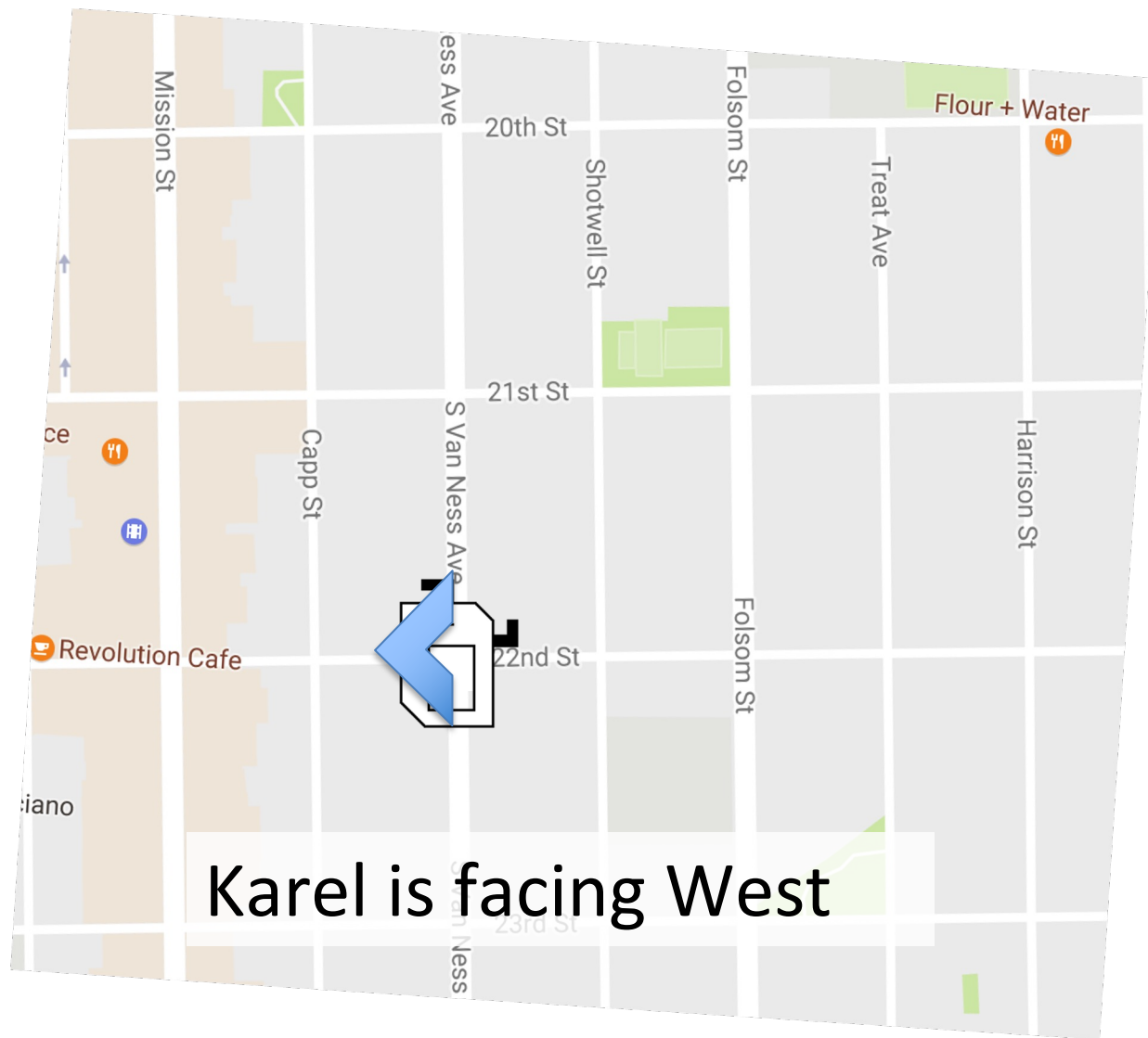
Bird's Eye View



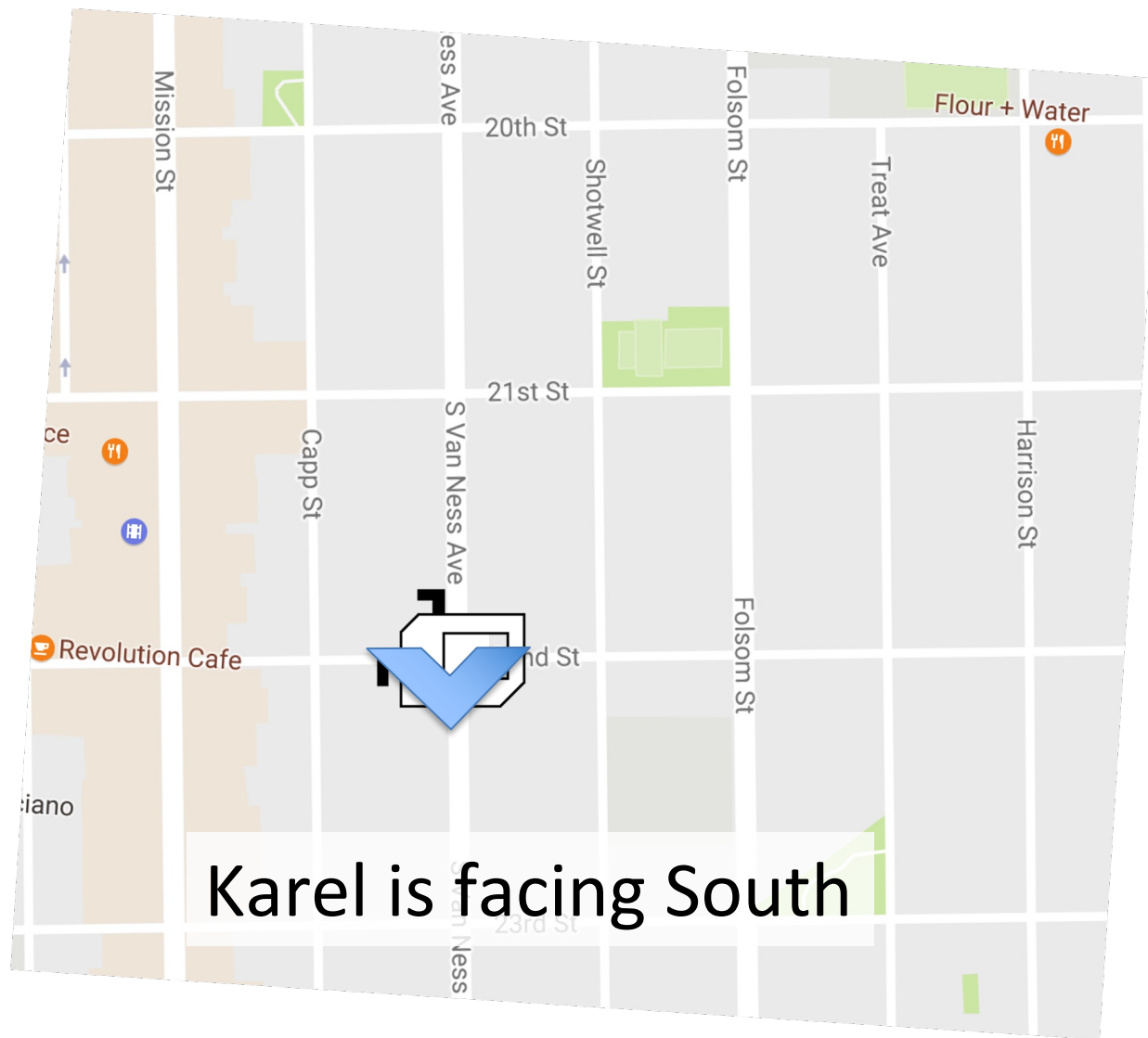
Turn Left



Turn Left

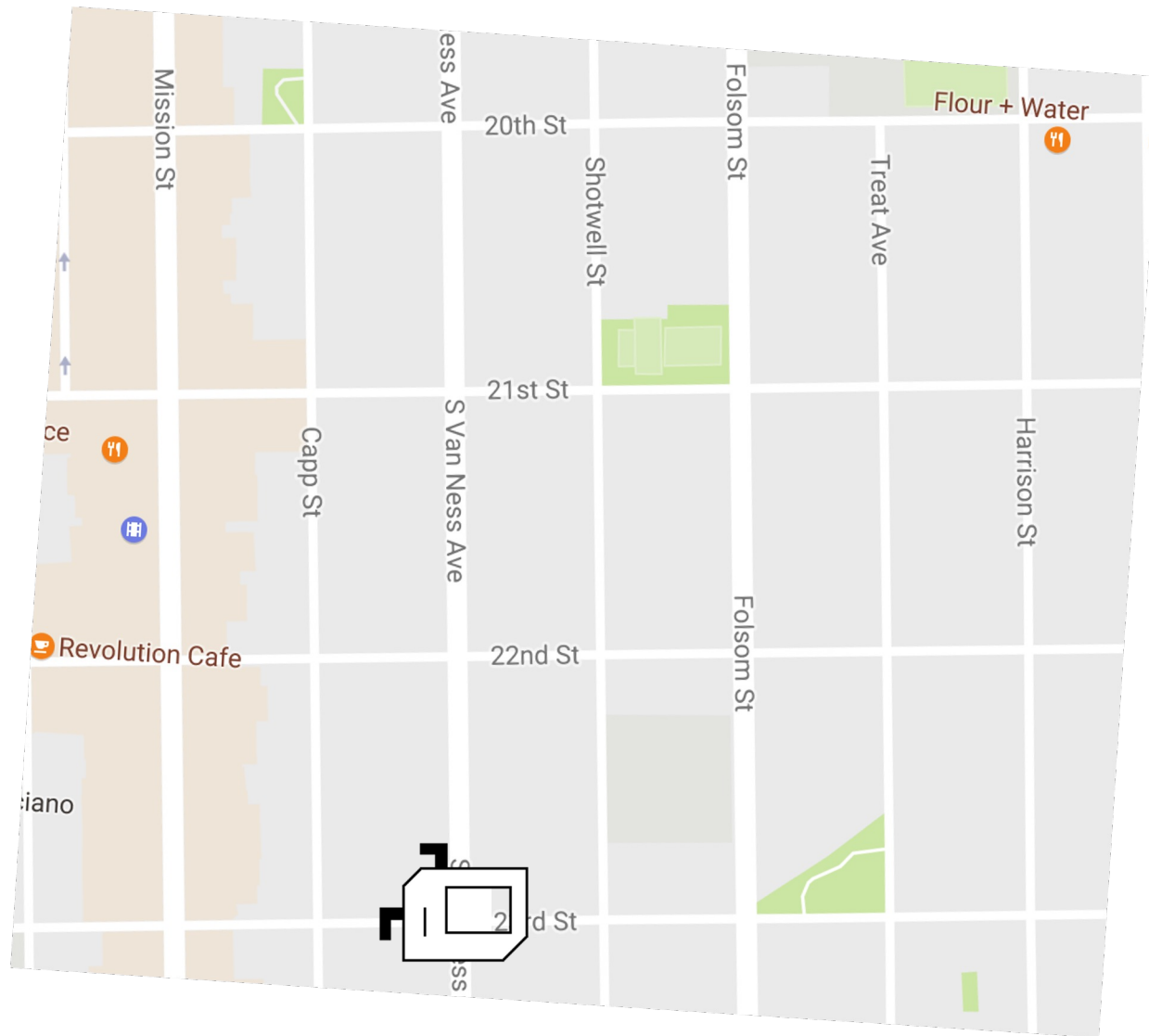


Turn Left



Karel is facing South

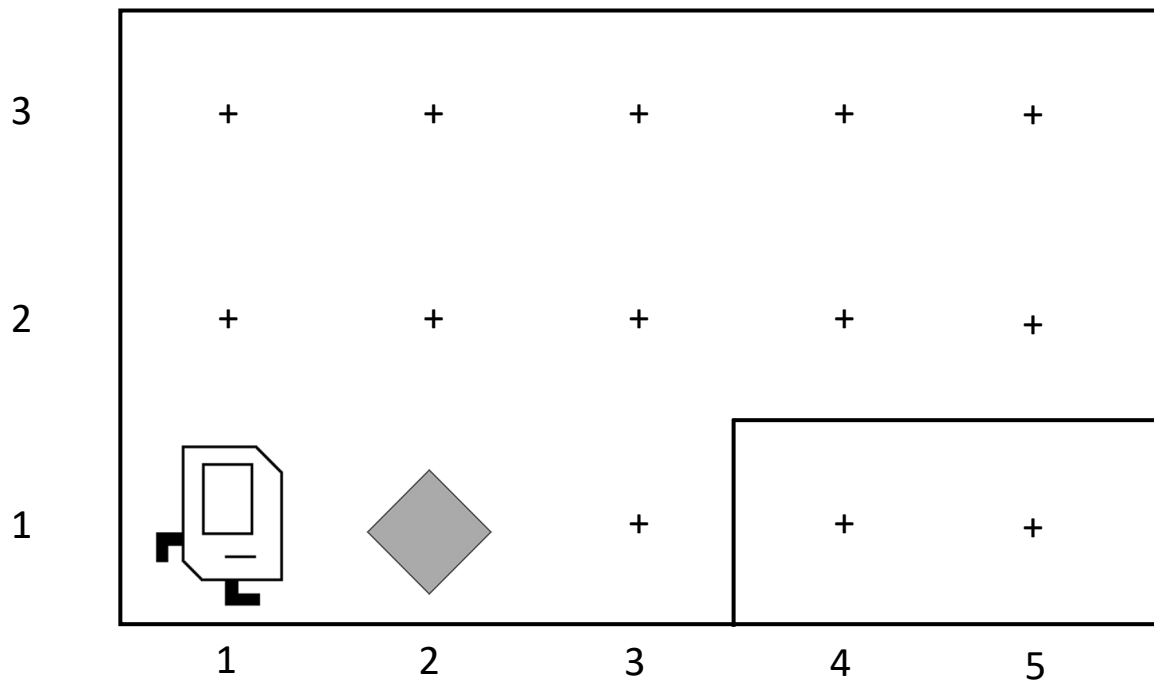
Move



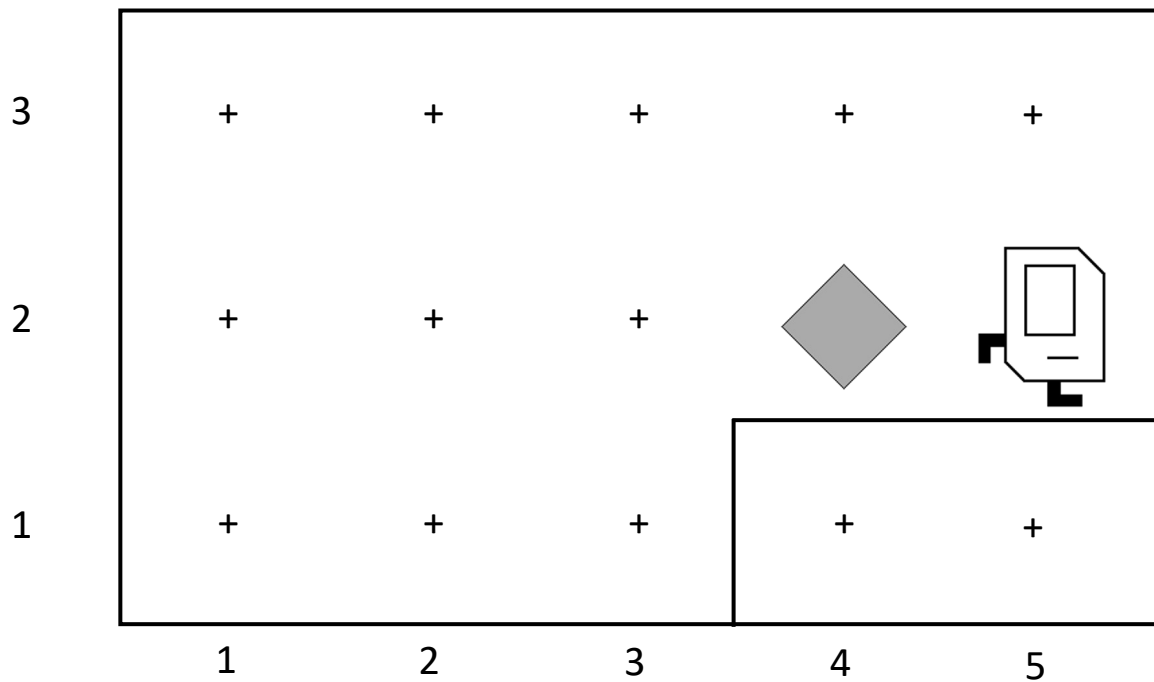
Piech and Sahami, Code in Place



First Challenge



First Challenge





Midjourney AI

Learn By Doing





The Python IDE
for Professional
Developers

DOWNLOAD

Full-fledged Professional or Free Community



Function Definition

```
def name() :  
    function statements
```

This adds a new
command to Karel's
vocabulary



Anatomy of a Program

Import Packages

Program



Anatomy of a Program

Import Packages



Anatomy of a Program

Import Packages

main function

helper functions

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

helper functions

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()  
  
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()  
  
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()  
  
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This piece of the program's **source code** is called a **function**.

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This line of code gives the ***name*** of the function (here, the name is: **main**)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```

This line of code gives the **name** of the function
(here, the name is: **turn_right**)



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():
```

```
    move()
    pick_beeper()
    move()
    turn_left()
    move()
    turn_right()
    move()
    put_beeper()
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():
```

```
    turn_left()
    turn_left()
    turn_left()
```

```
if __name__ == "__main__":
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```

